CS 116 - 03
Lab Assignment 6
Due: 03/14/2026

Please download the $CourseGrade.java$ and $Transcrpit.java$ files attached in the assignment and finish the implementation of the $Transcript$ class. Submit your implemented $Transcript.java$ to the Canvas.

Problem Description

In this lab assignment, we will be working with a class with an "array-of-objects" as attribute.

From the attached java files, we can see that a $CourseGrade$ object contains three attributes: a String $semester$, a String $courseNumber$ and a **char** $letterGrade$. There are some methods provided in the $CourseGrade$ class: a constructor, some accessor and mutator methods.

A $Transcript$ object has two attributes: an int $studentID$ and an array of $CourseGrade$ objects. There is only one constructor method in the $Transcript$ class now: it can construct a $Transcript$ object with a $studentID$ and an "empty" array (named $grades$) of $CourseGrade$ objects with length 100. Right after the construction of a $Transcript$ object, since no $CourseGrade$ objects are added in the array $grades$, Java simply uses 100 null objects to occupy 100 spots in the array:

$grades =$

null	null	null	null	null	null	null	null	null	...

If we add three $CourseGrade$ objects cg1, cg2 and cg3 to the array, $grades$ can look like this:

$grades =$

cg1	cg2	cg3	null	null	null	null	null	null	...

Note that, in this assignment, while adding a $CourseGrade$ object to the array $grades$, we always add it to the leftmost available spot in the array $grades$, where "available" means a spot occupied by a null object.

After adding objects cg1, cg2 and cg3, the length of the array $grades$ is still 100, we have that $(grades\ [0] == \text{null})$ is $False$, and $(grades\ [3] == \text{null})$ is $True$. We can call $grades\ [0].getSemester()$, but we cannot call $grades\ [3].getSemester()$, since $grades\ [3]$ is occupied by a null object.

Assignment

$Q1$ Implement in the $Transcript$ class the 4 following methods. Do not change any code in the main method. $(100\ pts)$

- `public boolean addCourseGrade(CourseGrade grade)`, which adds the **deep copy** of a $CourseGrade$ object `grade` to the corresponding $Transcript$ object. It returns $true$ if and only if the $CourseGrade$ object is successfully added. When the array $grades$ is full, then we cannot add more $CourseGrade$ objects into the $Transcript$ and the method returns $false$.
  - Remember that, in this assignment we always add a $CourseGrade$ object to the leftmost available spot of the $grades$ array. You can linear search for the leftmost available spot.
  - Note that there is no method in the $CourseGrade$ class that allows users to create deep copies; so, you need to create a deep copy by calling the constructor method and accessor methods here.

- **`public`** `CourseGrade[] getCourseGrade(String sem)`, which returns an array of **deep copies** of the $CourseGrade$ objects for the given semester `sem` in the corresponding $Transcript$. The output array should not contain any null objects.
  - If the corresponding $Transcript$ object contains no $CourseGrade$ objects, or contains no $CourseGrade$ objects in the given semester, then this method should output an array of length 0.

  For example, if a student has the following $CourseGrade$ objects in their $Transcript.grades$:

  ```
  2007F CS100 B
  2007F MATH151 A
  2007F CS201 D
  2008S CS330 I
  2008S MATH152 B
  ```

  Then, `getCourseGrade("2007F")` should output an array of length 3, with the following deep copies:

  ```
  2007F CS100 B
  2007F MATH151 A
  2007F CS201 D
  ```

- **`public double`** `getGPA()`, which calculates the GPA for the corresponding $Transcript$ as follows:
  - Each course worths 3 credit hours, and there are six different letter grades: A, B, C, D, E, I. Here, $gradePoints\ A = 4, B = 3, C = 2, D = 1, E = 0$.
  - Don't count any courses with grade $'I'$ (which represents incomplete) while calculating the GPA.
  - $\text{GPA} = [Summation\ of\ \{creditHours * gradePoints\}\ for\ each\ CourseGrades]$ $divided\ by\ [totalCreditHours]$
  - Return 0 if there are no credit hours.

  For example: If a student got one A, two B, one D and one I then his GPA is $\frac{(4+3+3+1)*3}{4*3} = 2.75$.

- **`public`** `String toString()`, which returns a String with the $studentID$, the total GPA, and list of String-representations of $CourseGrade$ objects in the corresponding $Transcript$ using the following format. Note that, keep at most two decimal places for a GPA (we can use `DecimalFormat("0.0#")` to format a GPA, and you can check previous assignments and lectures to see how to format a double.)

  ```
  SID = 1234 GPA = 2.75
  2007F CS100 B
  2007F MATH151 A
  2007F CS201 D
  2008S CS330 I
  2008S MATH152 B
  ```

Be careful of null objects while implementing all the above methods. Make sure to test your code on all the test cases provided. Check the test cases carefully for correct naming of methods and correct argument types.

Sample Output

```
Let's create a transcript for Jason, whose ID is 123456.
SID = 123456 GPA = 2.75
2020F CS331 I
2020F CS330 B
2020F MATH151 A
2021S CS425 I
2021S CS331 D
2021S MATH152 B
```

```
*********************************************************
Let's look at Jason's grades in 2021S.
SID = 123456 GPA = 2.0
2021S CS425 I
2021S CS331 D
2021S MATH152 B

*********************************************************
Let's test whether deep copy is used in the getCourseGrade method.
If deep copy is NOT used, the following code will change all Jason's grade to A.
SID = 123456 GPA = 2.75
2020F CS331 I
2020F CS330 B
2020F MATH151 A
2021S CS425 I
2021S CS331 D
2021S MATH152 B
```

As an aside, if deep copy was not used, then you will get:

```
*********************************************************
Let's test whether deep copy is used in the getCourseGrade method.
If deep copy is NOT used, the following code will change all Jason's grade to A.
SID = 123456 GPA = 4.0
2020F CS331 A
2020F CS330 A
2020F MATH151 A
2021S CS425 A
2021S CS331 A
2021S MATH152 A
```

Let’s continue with other test cases:

```
*********************************************************
Next, let's add more course grades to Jason's transcript.
SID = 123456 GPA = 2.67
2020F CS331 I
2020F CS330 B
2020F MATH151 A
2021S CS425 I
2021S CS331 D
2021S MATH152 B
2021F CS425 E
2021F HUM390 A
2021F MATH251 B
2021F PSYC303 A
2021F CS430 C

*********************************************************
In the end, let's create a transcript for Jamie who hasn't completed any course.
SID = 654321 GPA = 0.0
2020F CS331 I
2020F CS330 I
2020F MATH151 I
2021S CS425 I
2021S CS331 I
2021S MATH152 I
```